

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

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Windows PowerShell
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PS C:\Users\dhars> wsl
dhars@DHARSHAN: /mnt/c/Users/dhars$ cd /mnt/c/Users/dhars/OSP
dhars@DHARSHAN: /mnt/c/Users/dhars/OSP$ cd /mnt/c/Users/dhars/OSP/exec
dhars@DHARSHAN: /mnt/c/Users/dhars/OSP/exec$ nano program1.c
dhars@DHARSHAN: /mnt/c/Users/dhars/OSP/exec$ gcc program1.c -o program1
dhars@DHARSHAN: /mnt/c/Users/dhars/OSP/exec$ ./program1
Enter a Linux command: ls

Parent Process
Parent PID : 809
Child PID  : 810

Child Process
Child PID : 810
Parent PID: 809
Executing command...
program1 program1.c program2 program2.c

Child process completed.
dhars@DHARSHAN: /mnt/c/Users/dhars/OSP/exec$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:         39 bits physical, 48 bits virtual
Byte Order:            Little Endian
CPU(s):                12
On-line CPU(s) list:   0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-1355U
CPU family:            6
Model:                 186
Thread(s) per core:    2
Core(s) per socket:    6
Socket(s):              1
Stepping:               3
BogoMIPS:               5222.39
Flags:                  fpu vme de pse tsc mtrr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cpuid constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
```

```
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bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflush
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:          VT-x
Hypervisor vendor:       Microsoft
Virtualization type:     full
Caches (sum of all):
L1d:                      288 KiB (6 instances)
L1i:                      192 KiB (6 instances)
L2:                        7.5 MiB (6 instances)
L3:                       12 MiB (1 instance)
NUMA:
NUMA node(s):             1
NUMA node0 CPU(s):       0-11
Vulnerabilities:
Gather data sampling:     Not affected
Ghostwrite:              Not affected
Indirect target selection: Not affected
Itlb multihit:           Not affected
L1tf:                    Not affected
Mds:                     Not affected
Meltdown:                Not affected
Mmio stale data:         Not affected
Old microcode:           Not affected
Reg file data sampling:   Vulnerable: No microcode
Retbleed:                Mitigation; Enhanced IBRS
Spec rstack overflow:     Not affected
Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:              Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:              Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds:                   Not affected
Tsx:                     Not affected
Tsx async abort:         Not affected
Vmscape:                 Not affected
dhars@DHARSHAN: /mnt/c/Users/dhars/OSP/exec$ uname -a
Linux DHARSHAN 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
dhars@DHARSHAN: /mnt/c/Users/dhars/OSP/exec$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda  8:0  0 356.9M 1 disk
sdb  8:16 0 159.4M 1 disk
sdc  8:32 0 2G 0 disk [SWAP]
sdd  8:48 0 1T 0 disk /mnt/wslg/distro
```

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```

```
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ ps
PID TTY TIME CMD
735 pts/0 00:00:00 bash
887 pts/0 00:00:00 ps
```

```
dhars@DHARSHAN:/mnt/c/Users/dhars/OSP/exec$ top
top - 03:28:01 up 18 min, 1 user, load average: 0.14, 0.07, 0.01
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7782.6 total, 7082.6 free, 563.4 used, 287.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7219.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
897	dhars	20	0	9188	7108	3900	R	0.3	0.1	0:00.11	top
1	root	20	0	24080	15372	11592	S	0.0	0.2	0:00.50	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3180	2040	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	58396	16800	15528	S	0.0	0.2	0:00.16	systemd-journal
78	systemd+	20	0	22540	14600	11984	S	0.0	0.2	0:00.05	systemd-resolve
87	root	20	0	35364	12280	9188	S	0.0	0.2	0:00.48	systemd-udev
134	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.03	chronyd-starter
144	root	20	0	4472	3128	2880	S	0.0	0.0	0:00.01	cron
146	message+	20	0	8796	5300	4628	S	0.0	0.1	0:00.03	dbus-daemon
150	root	20	0	44092	29832	14696	S	0.0	0.4	0:00.20	networkd-dispat
154	root	20	0	18432	9388	8144	S	0.0	0.1	0:00.05	systemd-logind
174	syslog	20	0	220548	5732	4440	S	0.0	0.1	0:00.07	rsyslogd
203	root	20	0	5492	2748	2564	S	0.0	0.0	0:00.00	agetty
210	_chrony	20	0	20420	6444	5668	S	0.0	0.1	0:00.04	chronyd
212	root	20	0	123456	32544	17928	S	0.0	0.4	0:00.15	unattended-upgr
234	_chrony	20	0	12092	2460	1820	S	0.0	0.0	0:00.00	chronyd
298	root	20	0	8648	5536	4692	S	0.0	0.1	0:00.00	login
360	dhars	20	0	22508	13352	10900	S	0.0	0.2	0:00.08	systemd
362	dhars	20	0	22908	4108	2108	S	0.0	0.1	0:00.00	(sd-pam)
389	dhars	20	0	6136	5408	3860	S	0.0	0.1	0:00.02	bash
733	root	20	0	3184	1120	980	S	0.0	0.0	0:00.00	SessionLeader
734	root	20	0	3200	1260	1100	S	0.0	0.0	0:00.03	Relay(733)